Shot Peening Coverage Simulator

Model description, random laws and treatment
of the shot size distribution

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Developed with the assistance of Claude Opus 5.5 (Anthropic), October 2026

Abstract

The simulator is a self-contained web page that illustrates and checks the coverage model of the spreadsheet `tauxderecouvrement.xlsx`. A jet sweeps a part at traverse speed V; in a small observed area at the centre of the part, each shot leaves a randomly placed circular dent, and the surface actually covered is measured and compared with the law $\tau(n) = 100\,(1 - e^{-Kn})$. This note describes (i) the extension of the model to a log-normal distribution of shot diameters and dent radii, (ii) the random laws and the algorithm used, (iii) how to read the interface (map of the number of impacts per point, histogram, coverage curve) and (iv) a statistical validation over 20 independent simulations.

1 Main result

With a shot size distribution of spread σ (log-normal law) and dents proportional to the shot diameter ($r = c\,D$), the coverage law keeps the same form:

$$\tau(n) = 100\left(1 - e^{-Kn}\right), \qquad K = \frac{6\,\dot{m}\,c^2}{\rho\,D_{32}\,V\,\Delta X}, \qquad D_{32} = \frac{\mathbb{E}[D^3]}{\mathbb{E}[D^2]}. \tag{1}$$

D_{32} is the Sauter mean diameter. For a log-normal law with mass median D_{50}:

$$D_{32} = D_{50}\,e^{-\sigma^2/2} \quad \Longrightarrow \quad K(\sigma) = K_{\text{mono}}\,e^{\sigma^2/2}, \tag{2}$$

where K_{mono} is the spreadsheet value (all shots of diameter D_{50}). **The spreadsheet therefore remains valid when $D = D_{32}$ and $r = c\,D_{32}$ are entered.**

The size spread slightly *increases* coverage: at equal mass flow rate, a wide distribution contains many more small shots, hence more impacts, which compensates for their smaller dents.

2 Input parameters

Table 1: Adjustable parameters of the simulator (defaults = spreadsheet values).

Parameter	Symbol	Default	Range	Remark
Media flow rate	$\dot{m}$	1.5 kg/min	0.1 – 6	B2
Media density	ρ	2.2 g/cm^3	1 – 8	B4
Mass median diameter	D_{50}	150 µm	30 – 1000	sieving; B6 if $\sigma = 0$
Size spread	σ	0	0 – 0.6	std. dev. of $\ln D$
Jet size	ΔX	20 mm	5 – 60	B11; line pitch
Traverse speed	V	300 mm/s	20 – 1500	B12
Dent radius (for $D = D_{50}$)	r_{50}	10 µm	2 – 100	B16; $c = r_{50}/D_{50}$
Number of passes	$n_{\max}$	15	1 – 40	–

Typical values of σ: 0.1 to 0.2 for well-graded media, 0.3 to 0.5 for a wide size distribution or worn media (presence of fines).

3 Analytical model extended to size distributions

3.1 Log-normal diameter distribution

The logarithm of the diameter follows a normal law. Two ways of counting must be distinguished:

- **by number** (each shot counts as 1): median $D_{50,n}$;
- **by mass** (each shot counts by its mass $\propto D^3$): median D_{50}, the one given by sieving.

For a log-normal law, both distributions have the same σ and their medians are related by

$$D_{50,n} = D_{50}\, e^{-3\sigma^2}. \tag{3}$$

The number-weighted moments are $\mathbb{E}[D^k] = D_{50,n}^k\, e^{k^2\sigma^2/2}$, hence:

$$\mathbb{E}[D^3] = D_{50}^3\, e^{-4.5\,\sigma^2}, \qquad \mathbb{E}[D^2] = D_{50}^2\, e^{-4\,\sigma^2}, \qquad D_{32} = \frac{\mathbb{E}[D^3]}{\mathbb{E}[D^2]} = D_{50}\, e^{-\sigma^2/2}. \tag{4}$$

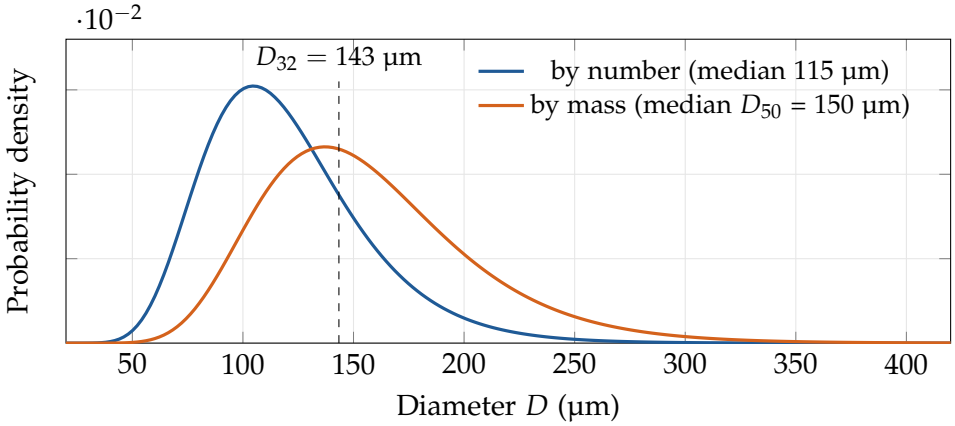


Figure 1: Log-normal distributions by number and by mass for $D_{50} = 150$ µm and $\sigma = 0.3$. Counting by number is dominated by fines, counting by mass by the large shots.

3.2 Dent radius proportional to the diameter

For a given impact velocity and target material, the dent is to a first approximation geometrically similar to the shot: $r = c\,D$. An energy balance (kinetic energy $\frac{1}{2}mv^2$ dissipated in the dent volume at hardness H) gives the order of magnitude

$$c \approx \left(\frac{\rho\, v^2}{6\, H}\right)^{1/4}, \tag{5}$$

independent of D. In the simulator, c is set by the user through the radius r_{50} measured (or estimated) for a shot of diameter D_{50}: $c = r_{50}/D_{50}$ ($= 10/150 \approx 0.067$ by default).

3.3 Calculation chain

$$\bar{m}_p = \rho\,\frac{\pi}{6}\,\mathbb{E}[D^3] \qquad \text{mean shot mass} \tag{6}$$

$$\dot{N} = \dot{m}/\bar{m}_p \qquad \text{shots thrown per second} \tag{7}$$

$$R = \dot{N}/(V\,\Delta X) \qquad \text{impacts per mm}^2 \text{ per pass} \tag{8}$$

$$K = \pi\,\mathbb{E}[r^2]\,R = \pi\,c^2\,\mathbb{E}[D^2]\,R \qquad \text{dent area deposited per unit surface per pass} \tag{9}$$

Substituting gives equation (1). Two points deserve attention:

- the number of shots depends on $\mathbb{E}[D^3]$ (mass) and the covered area on $\mathbb{E}[D^2]$ (area): their ratio D_{32} drives K;
- $\mathbb{E}[r^2] \neq (\mathbb{E}[r])^2$: using the mean radius underestimates the deposited area.

Table 2: Effect of the spread σ with the other spreadsheet parameters ($D_{50} = 150$ µm, $r_{50} = 10$ µm).

σ	$D_{50,n}$ (µm)	D_{32} (µm)	95 % range of r (µm)	$e^{\sigma^2/2}$	K	n_{98}
0.0	150	150.0	10.0–10.0	1.000	0.337	11.6
0.1	146	149.3	7.9–11.9	1.005	0.338	11.6
0.2	133	147.0	5.9–13.2	1.020	0.344	11.4
0.3	115	143.4	4.2–13.9	1.046	0.352	11.1
0.4	93	138.5	2.8–13.8	1.083	0.365	10.7
0.5	71	132.4	1.7–12.8	1.133	0.382	10.3
0.6	51	125.3	1.0–11.3	1.197	0.403	9.7

The "95 % range of r" column gives the interval containing 95 % of the dents (by number): $c\,D_{50,n}\,e^{\pm 2\sigma}$. The number of passes for 98 % coverage is $n_{98} = \ln 50/K$. The effect remains small (< 5 %) as long as $\sigma \leq 0.3$.

3.4 Why the exponential law remains exact

Impact centres form a homogeneous Poisson point process of density R per pass, and each impact carries a radius drawn independently of its position (*Boolean model*). A given point is covered by a dent of radius r if its centre falls within the disc of radius r around the point. The number of dents covering this point after n passes therefore follows a Poisson law with parameter

$$\lambda = n\,R\,\mathbb{E}[\pi r^2] = K\,n, \qquad P(k) = e^{-\lambda}\,\frac{\lambda^k}{k!}. \tag{10}$$

The unimpacted surface is $P(0) = e^{-Kn}$, hence $\tau = 100(1 - e^{-Kn})$ whatever the distribution of radii.

3.5 Use with a measured size distribution

If sieving gives mass fractions w_i for diameter classes D_i, the number of shots in each class is proportional to w_i / D_i^3 and

$$D_{32} = \frac{\sum_i w_i}{\sum_i w_i / D_i} \qquad \text{(Excel: } \texttt{=SUM(w)/SUMPRODUCT(w/D)}\text{).} \tag{11}$$

Then simply enter $D = D_{32}$ and $r = c\, D_{32}$ in the spreadsheet. If dent radii r_j are measured directly under the microscope, use $\mathbb{E}[r^2] = \overline{r_j^2}$ (mean of the squares) in $K = \pi\, \mathbb{E}[r^2]\, R$.

4 Simulation algorithm

4.1 Sweep geometry

The simulated part is a square of side $3\Delta X$ swept in three back-and-forth lines at pitch ΔX (figure 2). One *pass* is a complete sweep of the part. The observed area, a square of side L, is at the centre of the middle line. A point of this area stays under the jet for $\Delta X / V$ ($= 67$ ms by default); impacts appear in the area during this window. The "fast-forward outside the zone" option speeds up the animation while the jet is not over the area; it does not change the result.

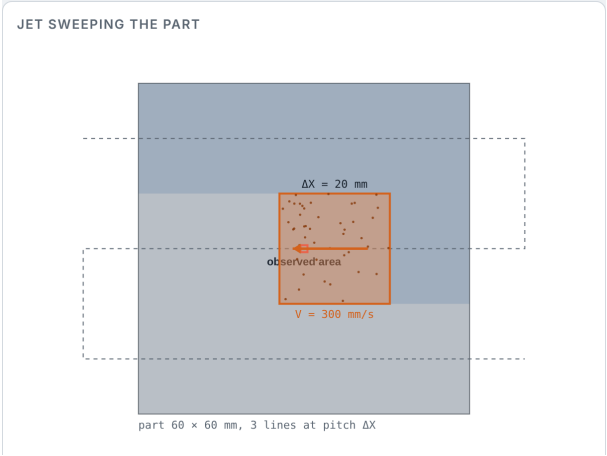


Figure 2: View of the part: the jet (orange square of side ΔX) sweeps the part at speed V; the bluish band is the surface already swept during the current pass; the small red square marks the observed area.

4.2 Size of the observed area

The area is chosen to contain a few hundred impacts per pass: $L = 50\, r_{\text{rms}}$, with $r_{\text{rms}} = \sqrt{\mathbb{E}[r^2]}$, reduced if necessary so that no more than 2 500 impacts are drawn per pass. With the default values, $L = 0.5$ mm and about 270 impacts land in the area at each pass.

4.3 Random draws

At each pass the simulator performs the following draws (the browser's pseudo-random generator, without a fixed seed):

1. **Number of impacts**: $N \sim \text{Poisson}\big(R\,(L + 2m)^2\big)$. Knuth's exact algorithm for a mean < 30, normal approximation $\mathcal{N}(\mu, \sqrt{\mu})$ rounded above (about 290 on average by default; negligible relative difference).

2. **Positions**: x, y uniform and independent on $[-m,\, L + m]$. The margin $m = c\, D_{50,n}\, e^{3.2\sigma}$ (largest likely dent) lets shots landing just outside the area overlap into it, which avoids an edge bias.

3. **Diameters**: $D_i = D_{50,n}\, e^{\sigma Z_i}$ with $Z_i \sim \mathcal{N}(0,1)$ (Box–Muller method); dent radius $r_i = c\, D_i$.

4. **Arrival time**: uniform during the passage of the jet over the area (visual effect only).

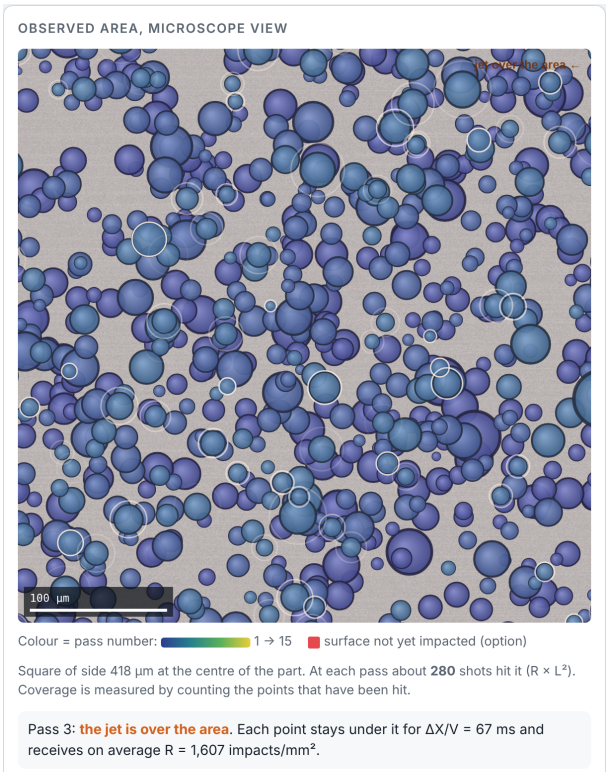
Positions, sizes and numbers are independent of each other and from one pass to the next: this is exactly the Boolean model of the previous section.

4.4 Measuring coverage and the number of impacts

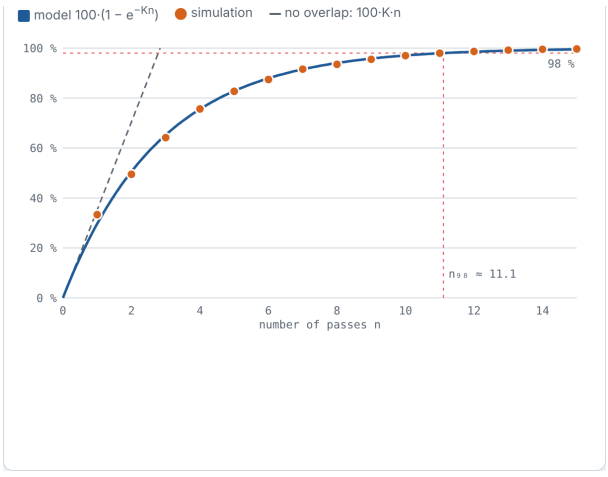
The area is discretised into a 500×500 grid of points (step $L/500$, i.e. 1 µm by default). Each impact increments a counter for every point whose distance to its centre is $\leq r_i$. This gives:

- the **measured coverage**: fraction of points with a non-zero counter;
- the **map of the number of impacts** per point and its **histogram**;
- the quantity $\lambda_{\text{meas}} = \sum_i \pi r_i^2 / (L + 2m)^2$ (total deposited area per unit surface), whose expected value is $K\,n$; the simulator also displays the predicted coverage $100(1 - e^{-\lambda_{\text{meas}}})$, which follows the measurement even in the middle of a pass.

5 Reading the interface



(a) Microscope view ($\sigma = 0.3$, pass 3). Colour = pass number.



(b) Curve $\tau(n)$: model (blue), simulation (dots), no-overlap case $100Kn$ (dashed).

Figure 3: Zoomed view and coverage curve.

Microscope view
Dents appear while the jet covers the area, coloured by pass (dark blue → yellow). The "show unimpacted surface" option colours the points not yet hit in red. The readouts below give n, the density of impacts received, λ_{meas}, the measured coverage and the predicted coverage.

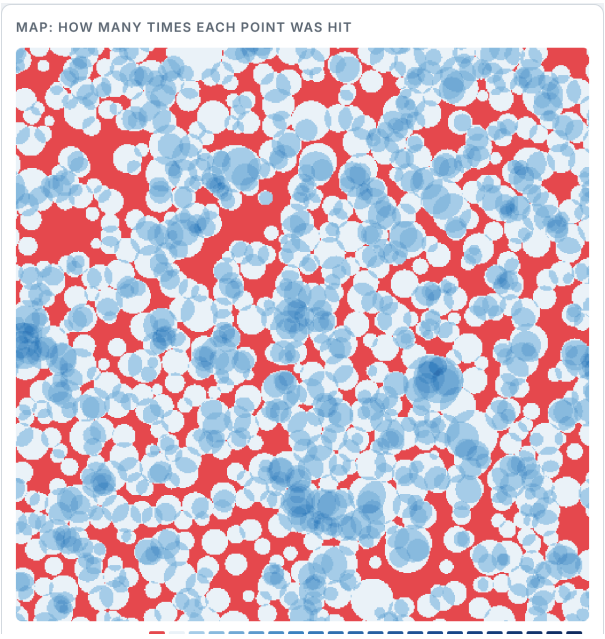
Coverage curve
The dashed line $100\,K\,n$ is the surface that would be covered without overlap; the gap with the blue curve is the effect of overlaps between dents.

Impact count map
Each point is coloured by the number of dents covering it: red = 0, then from light blue (1) to dark navy (≥ 25). Hovering gives the exact value at a point. At the end of the simulation, the "Map after n passes" slider recomputes the map for any intermediate pass.

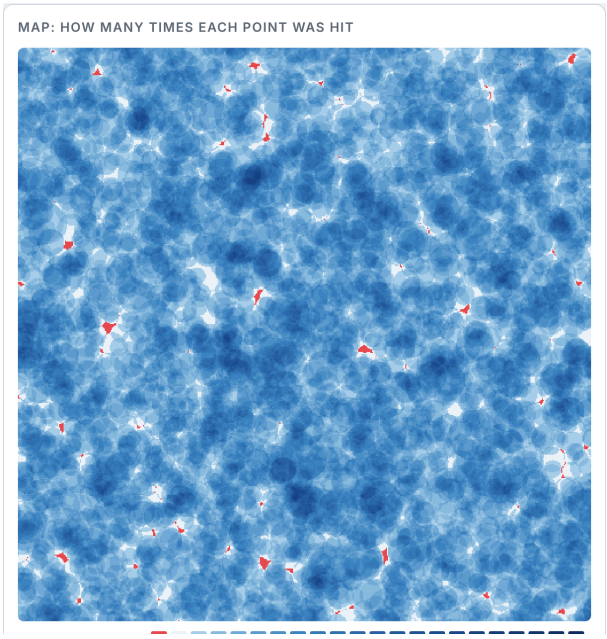
Histogram
Share of the surface covered exactly k times (bars, $k = 0$ in red) compared with the Poisson law (10) (blue dots).

Shot size distribution
Curves by number and by mass, with the markers D_{50} (mass), $D_{50,n}$ (number) and D_{32}, and the values to enter in the spreadsheet.

Calculation chain
The five steps of section 3.3, with values updated whenever a parameter changes.



(a) Map after 4 passes: about a quarter of the surface is still unimpacted (red).

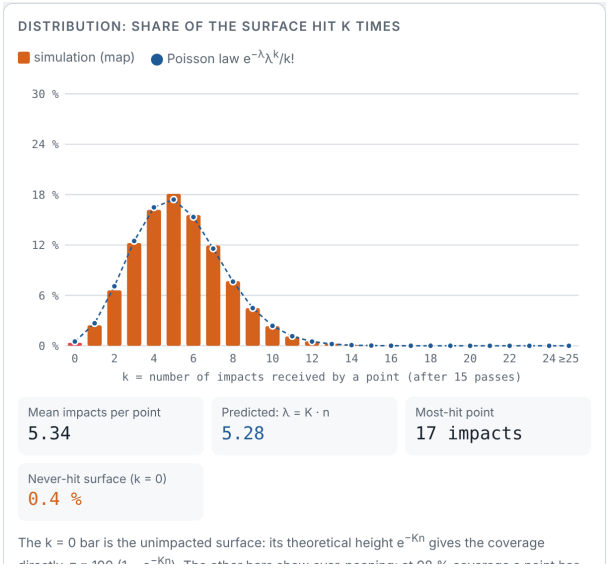


(b) Map after 15 passes: only a few unimpacted islands remain.

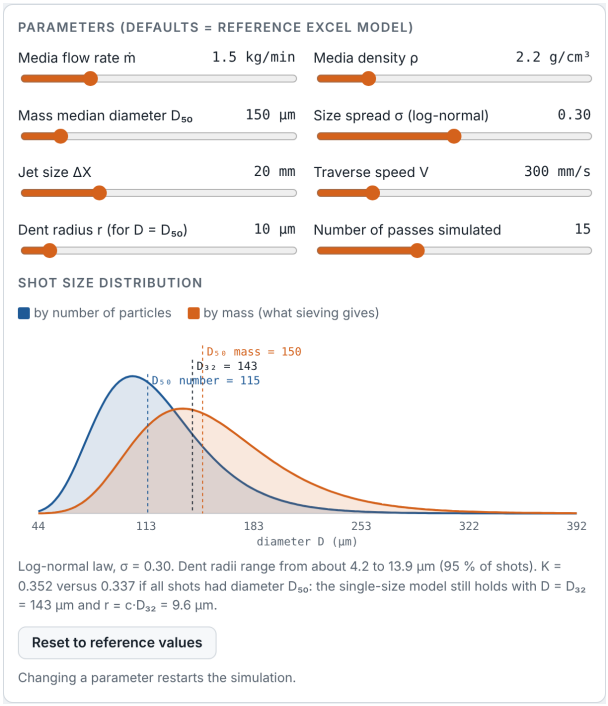
Figure 4: Evolution of the map of the number of impacts per point ($\sigma = 0.3$).

6 Statistical validation

The algorithm was rerun 20 times independently for $\sigma = 0$, 0.2 and 0.4 (250×250 grid, other parameters from the spreadsheet). Table 3 compares the mean of the 20 measured coverages with the law $100(1 - e^{-Kn})$, with K given by (2).



(a) Histogram after 15 passes and indicators.



(b) Parameters and shot size distribution.

Figure 5: Histogram of the number of impacts and parameter panel.

Table 3: Mean simulated / predicted coverage (%), mean of 20 simulations.

σ	K	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 12$	$n = 15$
0	0.337	28.8 / 28.6	49.1 / 49.0	73.7 / 74.0	93.0 / 93.2	98.2 / 98.2	99.3 / 99.4
0.2	0.344	29.3 / 29.1	50.2 / 49.7	75.2 / 74.7	93.7 / 93.6	98.4 / 98.4	99.5 / 99.4
0.4	0.365	30.8 / 30.6	51.8 / 51.8	76.7 / 76.8	94.6 / 94.6	98.7 / 98.7	99.6 / 99.6

Differences remain below 0.5 point, within the statistical margin. A single simulation fluctuates more: standard deviation of about 1.3 to 2.2 points at the first pass (0.5 mm area, a few hundred impacts), 0.2 point at 12 passes.

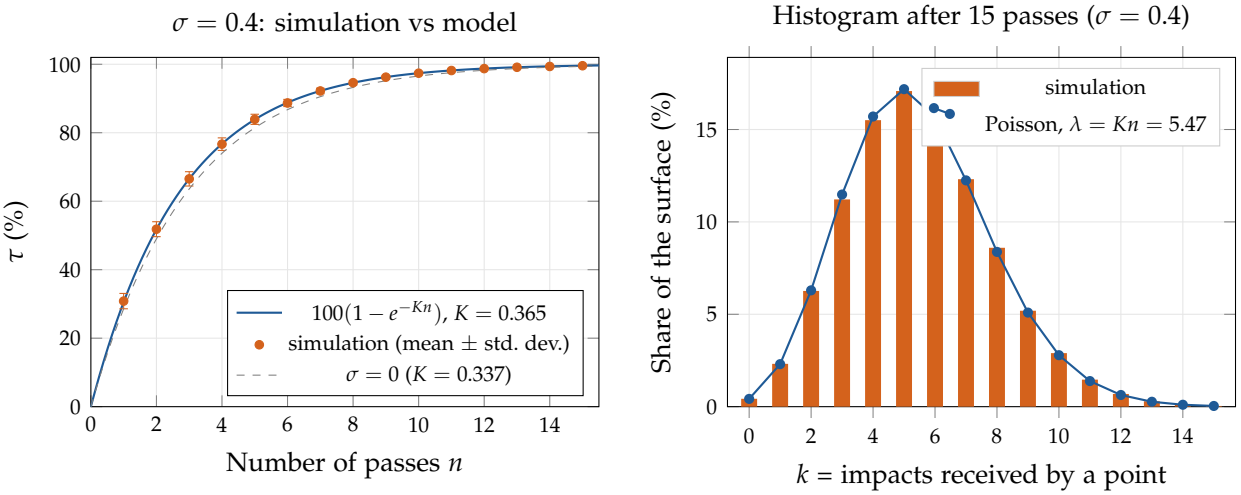


Figure 6: Validation for $\sigma = 0.4$ (mean of 20 simulations). Left: coverage curve; right: distribution of the number of impacts per point compared with the Poisson law.

7 Assumptions and limitations

1. **Uniform jet**: the impact density is the same across the whole jet width. A real, rather Gaussian profile covers the centre of the band better than its edges.
2. **Juxtaposed lines** at pitch ΔX, without overlap; for a pitch $p < \Delta X$, replace ΔX by p.
3. **Dent proportional to the shot** (c constant): assumes an impact velocity independent of size. If large shots are slower, use $c(D)$ per size class.
4. **Every shot marks**: no rebound, breakage or dent too faint to be visible; an efficiency $\eta < 1$ translates into $K \to \eta K$.
5. **Log-normal law**: convenient and realistic for many media, but a measured size distribution may deviate from it; in that case use $D_{32} = \sum w_i / \sum (w_i / D_i)$ directly.
6. **Flat surface, normal incidence**, no interaction between shots.

Correspondence with the spreadsheet

Excel	Simulator	Comment
B2, B4, B11, B12	$\dot{m}$, ρ, ΔX, V	identical
B6 (D)	D_{50} if $\sigma = 0$; otherwise enter $D_{32} = D_{50} e^{-\sigma^2/2}$	the simulator displays D_{32}
B16 (r)	r_{50} if $\sigma = 0$; otherwise enter $c\, D_{32}$	the simulator displays $c\, D_{32}$
B17 (K)	$K = \pi\, \mathbb{E}[r^2]\, R$	equal to the corrected spreadsheet value
G3:G42	blue curve	$100(1 - e^{-Kn})$

Dr Régis Kubler, Arts et Métiers Institute of Technology. Developed with the assistance of Claude Opus 5.5 (Anthropic), October 2026. Educational model with simplifying assumptions; it does not replace coverage verification according to the applicable standards (e.g. SAE J2277). Licence: CC BY 4.0.